



Riverside OS v0.85.5 — 30-Minute Quick-Start Deployment Guide

Authoritative reference manual for store hardware configuration, software package updates, schema migrations, and backup verification.

TARGET VERSION	v0.85.5
DOCUMENT CLASSIFICATION	Operational Standard Work
AUTHORIZED STATIONS	Server PC / Register #1 / Register #2 (iPad) / Counterpoint PC
GENERATED ON	June 3, 2026

CONFIDENTIAL • INTERNAL STORE OPERATIONS ONLY • DO NOT DISTRIBUTE

Table of Contents

- 30-Minute Timeline Overview
- Phase 1: Preparation (0-5 minutes)
- Phase 2: Server Install (5-12 minutes)
- Phase 3: Critical Integrations (12-18 minutes)
- Phase 4: Register #1 Setup (18-22 minutes)
- Phase 5: Register #2 iPad PWA (22-25 minutes)
- Phase 6: Test Sale and Go Live (25-30 minutes)
- Integration Troubleshooting Quick Reference
- Common Issues and Fixes
- Updates and Hotfixes
- Backup and Rollback
- DevOps Center (In-App)
- Additional Resources
- Post-Release Hotfixes (Applied to `main` after v0.85.5 tag)
- Deployment Log

Goal: Get fully up and running in 30 minutes (sync time not counted)

This guide takes you from zero to processing your first sale on Riverside OS. Print this guide and check off each step as you complete it.

30-Minute Timeline Overview

Phase	Time	What You'll Do
Preparation	0-5 min	Download files, gather info, prep network
Server Install	5-12 min	Install Backoffice/Server PC and verify
Integrations	12-18 min	Configure critical integrations (QBO, Helcim, etc.)
Register #1	18-22 min	Install main register with peripherals
Register #2	22-25 min	Set up iPad PWA satellite lane
Test Sale	25-30 min	Run smoke test and process first sale

Note: Initial data sync (Counterpoint, catalog, etc.) runs in background and doesn't count toward the 30 minutes.

Phase 1: Preparation (0-5 minutes)

What You Need Before Starting

Download Required Files:

- ☐ Download from: <https://github.com/cpg716/riverside-os/releases/tag/v0.85.5>
- ☐ RiversideOS-v0.85.5-Windows-Deployment.zip
- ☐ Riverside.POS_0.85.5_x64_en-US.msi
- ☐ Unzip the deployment package

Gather This Information:

- ☐ Backoffice/Server PC name and LAN IP (e.g., 10.64.70.196)
- ☐ Epson receipt printer IP or Windows printer name

- ☐ Scanner is in HID keyboard mode
- ☐ Payment terminal device codes (if using Helcim)
- ☐ Windows Administrator access on Server PC and Register #1

Network Setup:

- ☐ All devices (Server, Register #1, iPad) on same store network
- ☐ Confirm devices can reach each other
- ☐ Reserve static/DHCP IPs for server and printer (recommended)

Store Roles (Quick Reference):

Station	Device	API Host
Backoffice/Server PC	Windows PC	http://127.0.0.1:3000
Register #1	Windows PC	http://SERVER-IP:3000
Register #2	iPad PWA	http://SERVER-IP:3000

Critical Rule: Register #1 is the physical cash drawer lane. All other lanes attach to Register #1's till group.

Phase 2: Server Install (5-12 minutes)

Install Backoffice/Server PC

Step 1: Run Deployment Manager (2 minutes)

- ☐ On Server PC, open unzipped deployment package
- ☐ Run `Start-RiversideDeployment.cmd`
- ☐ Choose "Backoffice / Server"
- ☐ Click **Check** and resolve any failed checks
- ☐ Click **Install**

What the installer does:

- Installs/uses PostgreSQL
- Creates Riverside database and user
- Copies server files and web/PWA assets

- Applies database migrations
- Downloads ROSIE AI model (~5.4 GB, may take extra time)
- Sets up firewall rule for port 3000
- Creates auto-start task
- Starts Riverside Server

Step 2: Verify Server Health (3 minutes)

- ☐ Open <http://127.0.0.1:3000> on Server PC
- ☐ Staff sign-in screen loads
- ☐ Settings → Updates shows 0.85.5
- ☐ Settings → General → About shows correct API host
- ☐ Settings → Data & Backups opens
- ☐ Backup path is writable

Step 3: Test Network Access (2 minutes)

- ☐ From another device, open <http://SERVER-IP:3000>
- ☐ Staff sign-in screen loads
- ☐ No version mismatch warning

Server Repair Tools (if needed):

Tool	When to Use
Deployment Manager → Repair	Server files, firewall, task need refresh
Deployment Manager → PostgreSQL Status	Check/restart PG service
Repair-RiversideCredentialsKey.cmd	Settings says credentials key must be set
Set-CounterpointBridgeToken.cmd	Counterpoint bridge fails with 401

Phase 3: Critical Integrations (12-18 minutes)

Credential Key Setup (1 minute)

Required before saving any integration credentials:

- ☐ Sign in as administrator

- ☐ Navigate to Settings → any integration (e.g., Helcim)
- ☐ Try to save configuration
- ☐ If warning says `RIVERSIDE_CREDENTIALS_KEY` must be set :
 - ☐ Run `Repair-RiversideCredentialsKey.cmd` on Server PC (as admin)
 - ☐ Re-login and confirm integrations can be saved

QuickBooks Online (3 minutes)

For sales invoice and payment sync:

- ☐ Settings → Integrations → QuickBooks
- ☐ Enter Client ID and Client Secret
- ☐ Click "Connect to QuickBooks"
- ☐ Sign in with store QBO account, select Company file
- ☐ Authorize connection
- ☐ Configure account mappings:
 - ☐ Sales Income Account
 - ☐ Cash/Tender Deposit Bank Account
 - ☐ Sales Tax Liability Account
 - ☐ Accounts Receivable (AR) Account
 - ☐ Inventory Asset Account
 - ☐ COGS Account
- ☐ Verify QBO Sync Worker status is `Idle` or `Syncing`

Helcim Card Payments (3 minutes)

For physical card terminal integration:

- ☐ Settings → Integrations → Helcim
- ☐ Enter Merchant ID and API Access Token
- ☐ Set Environment to **Production** (unless testing)
- ☐ Configure terminal IP/device code for Register #1
- ☐ Trigger test ping, confirm "Terminal Connected"
- ☐ Copy webhook URL: `http://SERVER-IP:3000/api/webhooks/helcim`
- ☐ Configure in Helcim Developer Portal for events: `order.approved` , `transaction.captured`

Shippo Shipping (2 minutes)

For carrier rates and label printing:

- ☐ Settings → Integrations → Shippo
- ☐ Enter Shippo API Key (starts with `shippo_prod_` for live)
- ☐ Enter Default Store Origin Address
- ☐ Click Save
- ☐ Go to Fulfillment → Shipments Hub, verify carriers load

Meilisearch Search (2 minutes)

For fast product/customer search:

- ☐ Verify Meilisearch service running on port 7700
- ☐ Settings → Integrations → Meilisearch
- ☐ Confirm URL: `http://127.0.0.1:7700`
- ☐ Enter API Key
- ☐ Click "Test Connection" → status shows `Connected`
- ☐ Click "Full Reindex"
- ☐ Test search bar, confirm instant results

Optional Integrations (Configure as needed)

Podium Texting:

- ☐ Settings → Integrations → Podium
- ☐ Enter API Token and Location ID
- ☐ Verify webhook receiver is Active
- ☐ Map Staff Tracking IDs to Podium User IDs

Counterpoint Bridge:

- ☐ Settings → Integrations → Counterpoint
- ☐ Copy generated Bridge sync token
- ☐ On Counterpoint PC, set `RDS_BASE_URL=http://SERVER-IP:3000`
- ☐ Set `COUNTERPOINT_SYNC_TOKEN` to match
- ☐ Verify `CP_AUTO_SCHEMA` variable in bridge env:
 - If `CP_AUTO_SCHEMA=0` (manual-SQL mode), verify startup schema validation passes with no empty query check errors.
- ☐ Restart Counterpoint Node Bridge service
- ☐ Verify dashboard shows `Connected`

Sweden-Style Cash Rounding:

- ☐ Settings → Register (Terminal Overrides)
- ☐ Configure the "Cash Rounding" toggle (Sweden-style \$0.05 rounding)
- ☐ Verify that unauthenticated registers query the setting on lane initialization

NuORDER Wholesale:

- ☐ Settings → Integrations → NuORDER
- ☐ Save OAuth credentials (Consumer Key/Secret, User Token/Secret)
- ☐ Trigger health check, confirm `Connected`
- ☐ Link Vendors to Brands in Vendor Hub
- ☐ Trigger manual sync, verify no errors

Metabase Analytics:

- ☐ Verify Metabase container running on port 3001
- ☐ Connect database in Metabase (user: `metabase_ro`)
- ☐ Run metadata sync script
- ☐ Create collections: `Staff Approved` , `Admin Financial`
- ☐ Create user groups with appropriate permissions
- ☐ Test Insights iframe loads in Back Office

ROSIE AI Copilot:

- ☐ Confirm precompiled binaries exist: `llama-server.exe` , `sherpa-onnx-offline.exe` , and `sherpa-onnx-offline-tts.exe` inside `C:\RiversideOS\rosie\bin\`
- ☐ Confirm SenseVoice and Kokoro models copied to `C:\RiversideOS\rosie\stt\` and `C:\RiversideOS\rosie\tts\`
- ☐ Confirm Gemma model downloaded and verified: `C:\RiversideOS\rosie\models\gemma-4-e4b\google_gemma-4-E4B-it-Q4_K_M.gguf`
- ☐ Verify `rosie_ready` flag file is created at `C:\RiversideOS\rosie\rosie_ready`
- ☐ Verify `.env` contains `RIVERSIDE_LLAMA_MODEL_PATH` pointing to the Gemma GGUF model
- ☐ Test Help Center search with trial question (e.g. ROSIE chat)

Auxiliary APIs:

- ☐ Settings → Integrations → Fal.ai (visual AI)
- ☐ Settings → Integrations → Geoapify (delivery/mapping)
- ☐ Settings → Integrations → Weather (Visual Crossing)
- ☐ Settings → Integrations → Email (SMTP/IMAP)
- ☐ Settings → Integrations → Cloud Backups (S3)
- ☐ Settings → Online Store (JWT secret)

Constant Contact:

- ☐ Settings → Integrations → Constant Contact
 - ☐ Save OAuth Client ID and Secret
 - ☐ Click "Authorize Account", complete OAuth flow
 - ☐ Configure target mailing lists
 - ☐ Trigger manual sync, verify success logs
 - ☐ Configure webhook URL in Constant Contact portal
-

Phase 4: Register #1 Setup (18-22 minutes)

Install Register #1 Windows PC

Step 1: Run Deployment Manager (2 minutes)

- ☐ On Register #1, open unzipped deployment package
- ☐ Run `Start-RiversideDeployment.cmd`
- ☐ Choose "Register #1"
- ☐ Enter Server API base: `http://SERVER-IP:3000` (NOT `127.0.0.1`)
- ☐ Enter Register #1 station label
- ☐ Choose receipt printer mode (Network address recommended)
- ☐ Choose cash drawer setting
- ☐ Click **Check**, resolve any failed checks
- ☐ Click **Install**

What the installer does:

- Writes station config to `C:\ProgramData\RiversideOS\station-config.json`
- Installs Riverside desktop app
- Saves API base, printer, and cash drawer settings
- Launches the app

Step 2: First Launch Verification (2 minutes)

- ☐ Open Riverside desktop app
- ☐ Confirm app icon is Riverside logo
- ☐ Settings → Updates shows `0.85.5`
- ☐ Settings → General → About shows `http://SERVER-IP:3000`

- ☐ Staff sign-in works
- ☐ Open Register #1
- ☐ Enter opening float per store procedure
- ☐ Confirm Register #1 is cash drawer lane

Peripherals Setup (2 minutes)

Receipt Printer:

- ☐ Epson TM-m30III has paper
- ☐ Printer IP recorded (network mode) or Windows printer name (installed mode)
- ☐ Settings → Printers & Scanners saved
- ☐ Test receipt prints successfully
- ☐ PWA receipt test prints from Epson IP/port
- ☐ Receipt reprint path works

Cash Drawer & Rounding:

- ☐ Cash drawer connected to receipt printer
- ☐ Opens for CASH/CHECK tender
- ☐ Does NOT open for card-only tender
- ☐ Verify cash sales apply Sweden-style rounding to nearest \$0.05 when enabled
- ☐ Verify card, check, and gift card tenders use exact-cent precision
- ☐ Staff understand cash belongs to Register #1

Scanner:

- ☐ Scanner in HID keyboard mode
- ☐ Scanner sends Enter/Tab suffix (if desired)
- ☐ POS product search receives scanned text
- ☐ Inventory/search input receives scanned text

Reports and Tags:

- ☐ Report printing uses browser/Windows print
- ☐ Clothing tags use Zebra 2844 station
- ☐ Zebra network checks fail clearly if unreachable

Phase 5: Register #2 iPad PWA (22-25 minutes)

Prepare iPad (1 minute)

- ☐ iPad on same store Wi-Fi as Server PC
- ☐ iPad date/time correct
- ☐ Safari available
- ☐ Bluetooth scanner paired as keyboard/HID (if used)

Install PWA (2 minutes)

- ☐ Open Safari on iPad
- ☐ Visit: `http://SERVER-IP:3000`
- ☐ Confirm Riverside sign-in screen loads
- ☐ Tap **Share** → **Add to Home Screen**
- ☐ Name it: "Riverside Register 2"
- ☐ Launch from Home Screen icon (not Safari tab)

First Launch Verification (1 minute)

- ☐ Launch from Home Screen icon
- ☐ Staff sign-in works
- ☐ Settings → Updates shows `0.85.5`
- ☐ No "Update incomplete" warning
- ☐ Open Register #2 only after Register #1 is open
- ☐ Behaves as satellite lane
- ☐ Customer lookup works
- ☐ Item lookup/cart works
- ☐ Checkout drawer fits on iPad

iPad Scanner and Receipt (1 minute)

- ☐ Bluetooth scanner paired and working
- ☐ Scan barcode, text appears in focused field
- ☐ Server can reach network printer (if iPad printing expected)
- ☐ Receipt reprint from Register #1 if iPad print fails

Phase 6: Test Sale and Go Live (25-30 minutes)

Production Smoke Test (3 minutes)

Server:

- ☐ Reboot/restart Server PC
- ☐ "Riverside OS Server" starts automatically
- ☐ Sign-in works on Server PC
- ☐ Another device opens `http://SERVER-IP:3000`
- ☐ Backup path writable
- ☐ Deployment Manager PostgreSQL Status: running, OK

Register #1:

- ☐ Staff sign-in works
- ☐ Register #1 opens
- ☐ Scanner works
- ☐ Receipt prints
- ☐ Cash drawer opens for CASH/CHECK
- ☐ Card/Helcim flow validated (if enabled)
- ☐ Z-close path understood

Register #2 iPad:

- ☐ Launches from Home Screen icon
- ☐ Staff sign-in works
- ☐ Register #2 attaches after Register #1 open
- ☐ Scanner works (if used)
- ☐ Receipt fallback path understood
- ☐ No stale version warning

Process First Sale (2 minutes)

Run supervised test sale:

- ☐ Sign in as staff
- ☐ Open Register #1
- ☐ Search or scan an item
- ☐ Add item to cart
- ☐ Open checkout drawer

- ☐ Complete store-approved test sale
- ☐ Confirm receipt prints correctly
- ☐ Confirm cash drawer opens (if cash tender)
- ☐ Confirm transaction appears in history

Backup Before Go Live (1 minute)

- ☐ Create fresh database backup
- ☐ Confirm backup file exists with non-trivial size
- ☐ Keep backup off server PC if practical
- ☐ Record backup timestamp and filename
- ☐ Record who has rollback authority

You're Live! 🎉

Riverside OS v0.85.5 is now ready for production use.

Integration Troubleshooting Quick Reference

Integration	Issue	Solution
QBO	Shows Disconnected	Re-auth via "Connect to QuickBooks"
Helcim	Terminal times out	Verify terminal IP, confirm same subnet
Shippo	Carrier quote error 401	Replace invalid/revoked API key
Podium	Texts fail to deliver	Check Location ID, verify account credits
Meilisearch	Search returns zero	Confirm service running, click "Full Reindex"
Counterpoint	health 401 or 503	Run Set-CounterpointBridgeToken.cmd
NuORDER	Sync 401 unauthorized	Verify OAuth credentials match brand
Metabase	Insights iframe fails	Check container on port 3001, verify metabase_ro user
ROSIE	Help returns empty	Check LLM file path in .env
Constant Contact	Sync failed/oauth error	Verify Client ID/Secret, check redirect URI

Common Issues and Fixes

PWA will not load

1. Confirm device on correct network
2. Open `http://SERVER-IP:3000` in browser
3. Check DNS/server name
4. Check firewall on Server PC
5. Close/reopen Home Screen icon
6. If stale, remove icon, clear site data, re-add

Register app won't reach server

1. Check Settings → General → About for API base

2. Confirm Register #1 NOT pointed at 127.0.0.1
3. Open Server URL from Register #1
4. Run Repair from Deployment Manager

PostgreSQL down/unreachable

1. Open Deployment Manager
2. Check PostgreSQL Status panel
3. If stopped, click **Start PG**
4. If connection failed, verify PG installed and running
5. Use **Restart PG** if service running but connections failing

Receipt won't print

1. Check Printers & Scanners settings
2. For network mode, confirm Epson IP and port 9100
3. For installed mode, confirm Windows test page works
4. Restart desktop app after printer changes
5. For iPad, confirm server can reach network printer

Update incomplete

1. Update Backoffice/Server PC first
2. Update Register #1 desktop app
3. On iPad, Check app files and reload
4. Do not continue until all surfaces show 0.85.5

Updates and Hotfixes

After any update:

- ☐ Update Backoffice/Server PC first
- ☐ Confirm server starts, Settings shows 0.85.5
- ☐ Update Register #1 desktop app
- ☐ Relaunch Register #1, confirm Settings shows 0.85.5
- ☐ On iPad, Settings → Updates → Check app files
- ☐ If PWA update prompt appears, tap **Reload now** (when no sale in progress)
- ☐ Confirm no station shows **Update incomplete**

Never run sales if server, Windows app, and PWA show different versions.

Backup and Rollback

Before live use:

- ☐ Create fresh database backup
- ☐ Confirm backup file exists with non-trivial size
- ☐ Keep copy off server PC if practical
- ☐ Keep previous installer/build accessible
- ☐ Record backup timestamp and filename
- ☐ Record who has rollback authority

Rollback procedure:

1. Stop all registers
 2. Stop active checkouts
 3. Record reason for rollback
 4. Restore database backup before running older binaries
 5. Do not run older server against newer migrated data without approval
-

DevOps Center (In-App)

For staff with `ops.dev_center.view` permission.

Access:

1. Open Riverside on any device
2. Sign in as staff with DevOps permissions
3. Open Settings → ROS Dev Center

What it shows:

- Operational status (DB health, integrations, stations)
- Runtime diagnostics (memory, disk, service status)
- E2E health (latest CI test lane status)
- Alerts (open system alerts, ack/resolve actions)
- Stations (register heartbeat status and versions)

- Bug reports (pending bugs with server log snapshots)
- GitHub (recent workflow runs, releases, one-click builds)

Trigger release build (requires `ops.dev_center.actions` permission and `RIVERSIDE_GITHUB_TOKEN`):

1. In ROS Dev Center, scroll to GitHub DevOps
2. Select release version and click "Build Release"

Additional Resources

- **Full Deployment Guide:** See `docs/DEPLOYMENT_GUIDE.md` for detailed technical reference
- **Production Hardening:** See `docs/PRODUCTION_HARDENING_GUIDE.md` for enterprise features
- **Redis Integration:** See `docs/REDIS_INTEGRATION_GUIDE.md` for distributed caching
- **Job Queue:** See `docs/JOB_QUEUE_GUIDE.md` for background job processing
- **Metrics System:** See `docs/METRICS_SYSTEM_GUIDE.md` for business and technical KPIs

Post-Release Hotfixes (Applied to `main` after v0.85.5 tag)

The following bugs were identified during the v0.85.5 go-live and fixed in `main`. Pull a fresh copy of the deployment package from `main` or the next tag to include these fixes.

Fix 1 — PostgreSQL Console Password Prompt Loop (`install-server.ps1`, `repair-bootstrap-admin.ps1`)

Symptom: During the server install (or repair), a PowerShell console window opened and repeatedly prompted for a PostgreSQL password. Entering the correct password had no effect.

Root cause (three compounding bugs):

1. `Invoke-NativeCommand` did not redirect stdin — `psql` could capture the open console and block waiting for input when `PGPASSWORD` failed.
2. All auth-probe `psql` calls were missing the `-w` (`--no-password`) flag; without it `psql` falls back to interactive prompting instead of failing immediately.
3. `$env:PGPASSWORD = $null` does not remove the variable in PowerShell (it sets it to an empty string); replaced with `Remove-Item Env:\PGPASSWORD`.

Fix: `RedirectStandardInput = $true` + immediate `Close()` in `Invoke-NativeCommand`; `-w` added to all `psql` probe calls; `PGPASSWORD` cleanup changed to `Remove-Item Env:\PGPASSWORD`.

Fix 2 — ROSIE Binaries Missing; Empty LLM Model Path (`Install-RosieAiStack.ps1`)

Symptom: During server install, the output showed:

```
WARNING: Required ROSIE binary 'sherpa-onnx-offline.exe' is missing from the package
WARNING: ROSIE: LLM model is missing at ''. Skipping Riverside OS LLM Host scheduled task.
```

ROSIE showed as unavailable and `RIVERSIDE_LLAMA_MODEL_PATH` was empty in the server `.env`.

Root cause: The deployment ZIP never contains the sherpa-onnx binaries (large compiled executables are not in git). The old script threw a fatal exception when they were absent, causing `Install-RosieStack` to return `null` → empty model path → blank env var.

Fix: `Install-RosieAiStack.ps1` now auto-downloads all required components when not bundled:

- **sherpa-onnx v1.13.2** (Windows x64) from GitHub Releases → extracts `sherpa-onnx-offline.exe`, `sherpa-onnx-offline-tts.exe`, + DLLs
- **SenseVoice Small** STT model from HuggingFace
- **Kokoro-82M** TTS model from HuggingFace
- Gemma GGUF download failure is now a **warning** (not a crash) so STT/TTS work even if the model isn't downloaded yet.

Deployment Complete! 🎉

You now have Riverside OS v0.85.5 fully deployed and ready for production use. The system will handle initial data sync (Counterpoint, catalog, etc.) in the background.

Deployment Log

Item	Value
Deployment date	
Release version	0.85.5
Backoffice / Server PC name	
Backoffice / Server PC IP	
Server API host on server PC	http://127.0.0.1:3000
Register #1 PC name	
Register #1 API base	
iPad Register #2 URL	
Counterpoint PC name	
Counterpoint ROS_BASE_URL	
Epson receipt printer IP/name	
Zebra scanner model(s)	
Backup file name	
Backup timestamp	
Installer owner	
Store manager signoff	
Rollback authority	